

ARYAN BHAPKAR

Machine Learning • Robotics • Autonomous Systems

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Robotics and AI Engineer with an M.S. from Arizona State University and hands-on research experience at the Interactive Robotics Laboratory. Author of four peer-reviewed IEEE publications spanning computer vision, biometric authentication, and industrial robotics. Proven ability to develop and deploy sim-to-real machine learning systems, including a 5-DOF robotic arm achieving 96% real-world accuracy with 30ms latency. Strong expertise in deep learning, reinforcement learning, and ROS2, with a consistent track record of transforming research prototypes into reliable, production-ready systems.

EDUCATION

Arizona State University – Tempe, AZ	Aug 2024 - Dec 2025
Master of Science (M.S.) in Robotics and Autonomous Systems (Artificial Intelligence)	
NMIMS Mukesh Patel School of Technology Management & Engineering - Mumbai, India	Aug 2020 – May 2024
B. Tech in Artificial Intelligence	

TECHNICAL SKILLS AND CERTIFICATIONS

Machine Learning and AI: Deep Learning, Reinforcement Learning (TD3), CNNs, GANs, LLMs, Fine-Tuning, MLOps, NLP, Model Training, Data Preprocessing, Inference Optimization, Model Evaluation

Robotics & Control: ROS2, MuJoCo, URDF, Robot Kinematics, Trajectory Planning, Kalman Filtering, Dynamic Movement Primitives (DMPs), Inverse Kinematics (IK)

Computer Vision: OpenCV, YOLOv9, Optical Flow, Feature Extraction, Object Detection, Image Preprocessing, Visual Tracking

Programming: Python, SQL, Arduino C, R, Bash

Frameworks & Tools: NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, TensorFlow, AWS (EC2, Lambda, S3, DynamoDB), Matplotlib, Linux, MATLAB, ROS2 Gazebo

RESEARCH EXPERIENCE

Interactive Robotics Laboratory, Tempe, AZ:	Aug 2025 - Present
Research Aide (Advisors: Prof. Heni Ben Amor & Mohamed Mistiri)	
- Designed and deployed a high-fidelity MuJoCo simulation of a UR5e manipulator for adaptive tabletop manipulation, integrating reinforcement learning policies with Cartesian trajectory planning to enable robust contact-rich behavior - Architected an LLM-driven robot control stack using Google Gemini Flash 2.5 that converts free-form natural language commands into executable, real-time motion primitives closing the gap between human intent & embodied robot action.	

PROJECTS

5-DOF Robotic Arm Table Tennis Prototype	Spring 2025
- Engineered an end-to-end sim-to-real pipeline in Gazebo / ROS2 combining YOLOv9 ball detection, optical flow, Kalman filtering, and a TD3 reinforcement-learning policy for closed-loop trajectory control. - Achieved 95% real-world hit accuracy at 30ms decision-to-actuation latency, demonstrating successful zero-shot transfer of the trained policy from simulation to physical hardware.	
StyleGAN-Based Style Mapper	Fall 2023
Built a high-resolution, single-reference style-transfer system leveraging StyleGan, GAN inversion, and JojoGAN mapping arbitrary photographic inputs to user defined artistic styles with photo-realistic fidelity	

PEER – REVIEWED PUBLICATIONS (IEEE)

- **Mitigating the Effect of Shadows in Vision-Based Industrial Robot** - IEEE ICETA, 2024 ([DOI](#))
Shadow-resilient detection pipeline (lighting optimization + Patmax) deployed on the KUKA KR4 AGILUS for industrial environments
- **Color Sorting Pick Up and Drop Robotic Arm** - IEEE I2CT2 2024 ([DOI](#))
Sensor-driven color-sorting pick and place robotic arm for automated industrial sorting workflows.
- **EarSiamNet: Leveraging Siamese Networks & SVM for Ear Biometric Authentication** -IEEE MoSICOM 2023 ([DOI](#))
Siamese-network embeddings with HOG features and SVM classification – 97.86% authentication accuracy.
- **Open Set Masked Face Identification System** - IEEE ICCVMI Spring 2023 ([DOI](#))
Open-set identification system robust to face occlusion relevant to post-pandemic identity verification.